

SAHIL T CHAUDHARY

sahilchaudh@gmail.com | +1 (412) 696-6827 | [linkedin.com/in/sahiltc/](https://www.linkedin.com/in/sahiltc/) | sahilchaudhary.github.io

EDUCATION

Carnegie Mellon University

Master of Science in Mechanical Engineering – Research — **GPA: 4.0/4.0**

Pittsburgh, PA
May 2025

- **Coursework:** Planning and Decision-making, Introduction to Robot Learning, Optimal Control and Reinforcement Learning, Robot Localization and Mapping, Modern Control Theory, Machine Learning, Robot Dynamics and Analysis
- **Scholarship:** Fully funded Master's student
- **Course assistant** for Machine Learning and Artificial Intelligence for Engineers (24787)

Vellore Institute of Technology

Bachelor of Technology in Mechanical Engineering — **GPA: 9.05/10.0**

Vellore, India
May 2022

SKILLS

Programming Languages: C++, Python, Bash, MATLAB, Julia

Tools: Isaac Lab, Gazebo, Unity, PyTorch

Platforms and Software: ROS, Git, Linux, Docker, SolidWorks, Fusion 360, Ansys

WORK EXPERIENCE

Juxta Technology, Inc.

Robotics Engineer, AI

San Francisco, CA
September 2025 – Present

- Developed **learning-based, probabilistic, and filtering methods** for drift mitigation in **neural-inertial navigation and localization** using only IMU, achieving positioning accuracy with **~2 m ATE/RTE**
- Built a simulation framework for realistic human-motion trajectories and **synthetic noisy IMU measurements**, incorporating **diffusion** to bridge the **sim-to-real gap**; model trained on synthetic-only data achieves **~3 m ATE/RTE** on real-world testing
- Integrated **Kalman Filter**-based estimation with **neural-inertial odometry** and developed heuristic health monitoring and an **LSTM-based critic network** for estimator selection, improving tracking accuracy by up to **33%**
- Combined convolution-based trajectory initialization with **map matching** for global localization, developing hypothesis-pruning and state-selection methods across five candidate locations, achieving **61% success rate**
- Investigated heuristic methods for **yaw-drift correction** and **gyro-bias estimation**, evaluating robustness across trajectories and identifying failure modes that limited generalization

Biorobotics Lab, CMU Robotics Institute

Graduate Research Assistant (Paid during Summer 2024)

Pittsburgh, PA
August 2023 – May 2025

- Contributed to the implementation of an **Inverse Kinematics-based optimization** framework for optimal communication node placement
- Spearheaded the development of a **MANET framework** using DDS and ROS to ensure communication fidelity in heterogeneous robot convoys, perform network topology repair and recovery behaviors, and enforce a communication boundary
- **Comms-Aware Planning:** Designed a novel algorithm attaining **99% success rate** to maintain communication among robots over radio, by formulating a modified Max-Min Spanning Tree, and validated the algorithm through extensive hardware testing
- **Heterogeneous Convoy Framework:** Developed a decentralized convoy framework integrating RC cars and quadrupeds like Boston Dynamics' Spot, and algorithms enabling intersection rendezvous and coordinated return for exploration missions
- Enhanced the operational efficiency of the **local planner by up to 29%**, through waypoint optimization and trajectory smoothing, reducing unnecessary deceleration between waypoints and improving overall robot speed and motion continuity
- **Payload Redesign:** Engineered a modular, serviceable payload for RC cars and quadrupeds, achieving **7% weight reduction** and lower CG while optimizing **sensor FOV and accessibility** across LiDAR, IMU, cameras, compute, and electronics

PUBLICATIONS

Sahil Chaudhary*, Charles Noren*, Burhanuddin Shirose, Bhaskar Vundhurty, Matthew Travers, "Communication Network Construction Behaviors for Robotic Convoys" *Published in GVSETS 2025*

PROJECTS

Blind Manipulation [Website]

Search-Based Planning Lab, CMU RI – Research Project

Pittsburgh, PA
May 2025 – September 2025

- Trained a PPO agent in **Isaac Lab** for last-mile manipulation, achieving a **94% success rate**
- Developed a **hierarchical framework** for navigation and manipulation in **unknown environments using only contact sensing**

Quadruped Path Planner for Dynamic Environments [Website]

Carnegie Mellon University – Course Project

Pittsburgh, PA
September 2024 – November 2024

- Demonstrated a global path planner using C++ and ROS, that accounts for dynamic obstacles and the z-height of the robot
- Executed Lazy-PRM with D-Star Lite and kinodynamic constraints in Gazebo, achieving a **93% success rate**

Model Predictive Path Integral Control [Website]

Carnegie Mellon University – Course Project

Pittsburgh, PA
February 2024 – April 2024

- Implemented MPPI using C++ and ROS in simulation, with obstacle avoidance leveraging a Voxel Grid Costmap
- Benchmarked MPPI against an existing iLQR controller, demonstrating **faster path generation of up to 21**

Point-LiDAR Inertial Odometry [Website]

Carnegie Mellon University – Course Project

Pittsburgh, PA
February 2024 – April 2024

- Engineered the Point-LIO algorithm using C++, ROS and GTSAM, addressing the limitations of scan-based LiDAR processing
- Performed **EKF-based state estimation**, modeling IMU measurements in the state vector to address **IMU saturation**